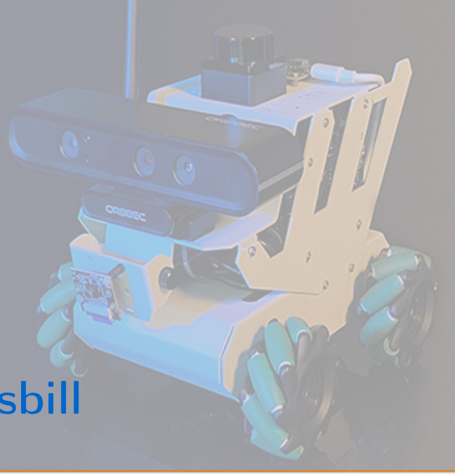

ROS Course

with ROS 2 Humble Hawksbill

Dr. Eduardo de Jesús Dávila Meza

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Session 1 | ROS 2 Environment Setup and Development Tools

Basic Concepts:

ROS (Robot Operating System) is a flexible *framework* that provides a set of software libraries and tools to build robot applications. From drivers and state-of-the-art algorithms to powerful developer tools, ROS has the open source tools you need for your next robotics project. Some of its main advantages include:

- **Multi-language:** Compatible with C, C++, Java, and Python.
- **Free and open-source:** Available in all Ubuntu versions (ROS 1 & 2), and for some Windows and MacOS distributions (ROS 2).
- **Support:** Since ROS was started in 2007, a lot has changed in the robotics and ROS community. The goal of the ROS 2 project, started in 2015, is to adapt to these changes, to take advantage of what is great about ROS 1 and to improve what is not.

Distributions

ROS is available for different versions of the Ubuntu distribution and other operating systems:

- **Indigo Igloo (ROS 1):** Ubuntu 14.04 (Trusty Tahr)
- **Kinetic Kame (ROS 1):** Ubuntu 16.04 (Xenial Xerus)
- **Melodic Morenia (ROS 1):** Ubuntu 18.04 (Bionic Beaver)
- **Noetic Ninjemys (ROS 1):** Ubuntu 20.04 (Focal Fossa)
- **Foxy Fitzroy (ROS 2):** Ubuntu 20.04 (Focal Fossa)
- **Humble Hawksbill (ROS 2):** Ubuntu 22.04 (Jammy Jellyfish)
- **Jazzy Jalisco (ROS 2):** Ubuntu 24.04 (Noble Numbat)
- **Kilted Kaiju (ROS 2):** Ubuntu 24.04 (Noble Numbat)

Distribution to be used for the course

Figure 1: ROS 2 Humble Hawksbill

Installing Ubuntu 22.04.5 LTS (Jammy Jellyfish)

1. **Download the Universal USB Installer:** Visit the Uptodown website to download the Universal USB Installer, version `2.0.1.4`, on Windows.
2. **Download the Ubuntu ISO:** Download the Ubuntu 22.04.5 LTS (Jammy Jellyfish) ISO - *64-bit PC (AMD64) desktop image*, from the official release page: Ubuntu releases.
3. **Create a Bootable USB Drive:** Use the Universal USB Installer tool with the downloaded ISO to create a bootable USB stick (minimum 8 GB). Follow the on-screen prompts to properly set up the drive.
4. **Prepare Your Machine (if dual-booting with Windows):** If you are keeping Windows and installing Ubuntu alongside it, you should free some unallocated disk space before installation: Defragment the `C:` drive and shrink it to create unallocated space for Ubuntu.
5. **Install Ubuntu on Your Machine:** Boot your computer from the USB drive. Follow the Ubuntu installation wizard to install Ubuntu 22.04.5 LTS and configure your system settings as needed.

Installing ROS 2 Humble

Follow the installation instructions on the official ROS 2 Humble Installation Guide. This method installs pre-built binaries and generally covers all core dependencies needed for running ROS 2.

Set locale

Make sure you have a locale which supports UTF-8. If not, use the next settings already tested.

```
$ locale  # check for UTF-8

$ sudo apt update && sudo apt install locales
$ sudo locale-gen en_US en_US.UTF-8
$ sudo update-locale LC_ALL=en_US.UTF-8 LANG=en_US.UTF-8
$ export LANG=en_US.UTF-8

$ locale  # verify settings
```

Installing ROS 2 Humble

Setup Sources

You will need to add the ROS 2 apt repository to your system. First, ensure that the Ubuntu Universe repository is enabled.

```
$ sudo apt install software-properties-common
$ sudo add-apt-repository universe
```

Now, add the ROS 2 GPG key with apt:

```
$ sudo apt update && sudo apt install curl -y
$ sudo curl -sSL
↪  https://raw.githubusercontent.com/ros/rosdistro/master/ros.key -o
↪  /usr/share/keyrings/ros-archive-keyring.gpg
```

Installing ROS 2 Humble

Setup Sources (continued)

Then, add the repository to your sources list:

```
$ echo "deb [arch=$(dpkg --print-architecture)
↪ signed-by=/usr/share/keyrings/ros-archive-keyring.gpg]
↪ http://packages.ros.org/ros2/ubuntu $(. /etc/os-release && echo
↪ $UBUNTU_CODENAME) main" | sudo tee /etc/apt/sources.list.d/ros2.list >
↪ /dev/null
```

Installing ROS 2 Humble

Install ROS 2 Packages

Update your apt repository caches after setting up the repositories:

```
$ sudo apt update
```

Ensure your system is up to date before installing new packages:

```
$ sudo apt upgrade
```

Desktop Install (Recommended): This includes ROS, RViz, demos, and tutorials.

```
$ sudo apt install ros-humble-desktop
```

Development Tools: Compilers and other tools to build ROS packages.

```
$ sudo apt install ros-dev-tools
```

Installing ROS 2 Humble

Environment Setup – Sourcing the Setup Script

After installing ROS 2, source the setup script to ensure your environment is correctly configured:

```
$ source /opt/ros/humble/setup.bash
```

Optionally, add the sourcing command to your shell's initialization file (e.g., `.bashrc`) so that the ROS environment variables are automatically loaded every time a new terminal session is opened (replace `.bash` with your shell if not using bash).

```
$ echo "source /opt/ros/humble/setup.bash" >> ~/.bashrc
$ source ~/.bashrc
```

Installing Visual Studio Code

To install VS Code on Ubuntu 22.04, follow these steps:

1. **Using the Ubuntu Software Center**
   1) Open the *Ubuntu Software* application from the applications menu.
   2) In the search bar, type *Visual Studio Code*.
   3) Locate *Visual Studio Code* in the search results and click *Install*.
2. **Using the Official .deb Package**
   1) Visit the official VS Code download page: `code.visualstudio.com`.
   2) Click on the *.deb* package suitable for Debian/Ubuntu.
   3) Once downloaded, open a terminal and navigate to the directory containing the downloaded file.
   4) Install the package using (replace `<file>` with the actual filename):

```
$ sudo apt install ./<file>.deb
```

Installing Recommended VS Code Extensions

Enhance your development experience by installing the following VS Code extensions:

- **C++** by *Microsoft* – Offers C++ IntelliSense, debugging, and code browsing.
- **CMake** by *twxs* – Simplifies working with CMake projects.
- **CMake Tools** by *Microsoft* – Provides CMake project integration.
- **Error Lens** by *Alexander* – Displays inline error messages in the code editor.
- **Indent-Rainbow** by *oderwat* – Highlights indentation levels with different colors.
- **Python** by *Microsoft* – Provides rich support for Python, including features such as IntelliSense, linting, and debugging.
- **Robotics Developer Environment** by *Ranch Hand Robotics LLC* – Helps build robotics solutions targeting the ROS 2.
- **vscode-icons** by *VSCode Icons Team* – Adds file icons for better visual identification.
- **XML** by *Red Hat* – Enhances XML editing capabilities.
- **XML Tools** by *Josh Johnson* – Offers additional functionalities for XML files.

Installing Terminator for Multiple Terminals

Terminator is a terminal emulator that allows splitting the window into multiple terminals, facilitating simultaneous operations. To install Terminator:

1. Open a terminal.
2. Update the package list:
```
$ sudo apt update
```
3. Install Terminator:
```
$ sudo apt install terminator
```
4. Launch Terminator from the applications menu or by pressing `Ctrl+Alt+T`.

With these tools and extensions, your development environment will be well-equipped for ROS 2 projects.

colcon Command

colcon is the command-line tool used in ROS 2 to build sets of packages in a workspace. It automatically detects packages (via package.xml), resolves dependencies, and builds them (often in parallel) to simplify the development process. colcon also supports options like -symlink-install for faster iterative development.

Installing colcon

Update your package list and ensure that the common colcon extensions are installed. Although these packages are usually installed as part of the ros-humble-desktop and ros-dev-tools installations, it is good practice to verify their presence by running the following commands:

```
$ sudo apt update
$ sudo apt install python3-colcon-common-extensions
```

Enabling colcon Argument Completion

To simplify command usage with colcon, add the argument completion script to your shell initialization file (e.g., .bashrc). Execute the following commands:

```
$ echo "source /usr/share/colcon_argcomplete/hook/colcon-argcomplete.bash"
↪  >> ~/.bashrc
$ source ~/.bashrc
```

You can verify that the sourcing command was added by viewing your .bashrc file:

```
$ cat ~/.bashrc
```

Talker-Listener

If you installed `ros-humble-desktop`, try some examples. In one terminal, source the setup file and run a C++ talker:

```
$ source /opt/ros/humble/setup.bash
$ ros2 run demo_nodes_cpp talker
```

In another terminal, source the setup file and run a Python listener:

```
$ source /opt/ros/humble/setup.bash
$ ros2 run demo_nodes_py listener
```

You should see the talker publishing messages and the listener confirming reception, verifying that both the C++ and Python APIs are working properly.

Turtlesim: Publish and Move a Turtle

First, ensure that the package is installed:

```
$ sudo apt install ros-humble-turtlesim
```

Then, in a new terminal, launch the turtlesim node:

```
$ ros2 run turtlesim turtlesim_node
```

Next, open another terminal, use the `teleop` node to control the turtle with your keyboard:

```
$ ros2 run turtlesim turtle_teleop_key
```

Now, pressing the arrow keys will move the turtle in the corresponding direction. This example demonstrates publishing velocity commands to a topic that the turtlesim node subscribes to, allowing interaction with the simulated environment.

Turtlesim: Publish and Move a Turtle (continued)

To get the turtle moving and keep it moving, you can use the following command. It is important to note that this argument needs to be input in the YAML syntax. Input the full command like so:

```
$ ros2 topic pub /turtle1/cmd_vel geometry_msgs/msg/Twist "{linear: {x:
↪  2.0, y: 0.0, z: 0.0}, angular: {x: 0.0, y: 0.0, z: 1.8}}"
```

Thank you for your attention!

Subsection

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